NETWORK INFORMATION THEORY 2026-2027.

1) It is well known that Gaussian is the unique entropy maximizing distribution under a variance constraint. Consider a AWGN MAC channel given by the receiver observations

$$Y = \alpha X_1 + \beta X_2 + Z$$

where Z is complex circularly symmetric Gaussian (c.c.s.g) of unit variance. The two transmitters have average power constraints of P_1 and P_2 respectively.

(a) Assuming fixed complex numbers α and β, argue that Y has to be Gaussian for achieving the sum-capacity of the above MAC .

(b) Propose a time sharing scheme where each user transmits for an appropriate fraction of the total available time, while the other stays silent, such that the received symbols are Gaussian distributed and the power constraints are met with equality. Comment whether time-sharing can achieve the sum-capacity of the above MAC system.

(c) Imagine a quazi-static fading system where the fading coefficients α and β stays constant for a sufficiently large block of transmissions, and varies across blocks in an ergodic fashion. If the marginal distributions of fading is normalized zero mean Gaussian, what do you think is the ergodic (long term average) sum-capacity of the above MAC channel.

(d) For two positive semi-definite matrices A and B, show that

$$\log\det(I + A + B) \leq \log\det(I + A) + \log\det(I + B) .$$

Hint: think of a Gaussian MAC with vector inputs, and connect to the capacity pentagon

2) Assume the entropy power inequality (EPI) for two scalar variables as follows :

$$2^{h(X+Y)} \geq 2^{h(X)} + 2^{h(Y)} ,$$

for independent complex valued X and Y . Note: h(.) is the differential entropy

(a) Show that $2^{h(X+Y|W)} \geq 2^{h(X|W)} + 2^{h(Y|W)}$, when X and Y are conditionally independent

given the third variable W. Feel free to refer to the textbook for hints.

(b) For n-length vectors $\bar{X}$ and $\bar{Y}$, show that

$$2^{\frac{1}{n}h(\bar{X}+\bar{Y}|W)} \geq 2^{\frac{1}{n}h(\bar{X}|W)} + 2^{\frac{1}{n}h(\bar{Y}|W)}$$

where the vectors are conditionally independent given the variable W.

Note: Holder's inequality is very handy for an iterative proof. You can also do the proof without the variable W first, and later put the conditioning back using ideas from part (a).